A Full DR11 Strong-Lens Search in the DESI Legacy Surveys: ClaudeNet `v3blend8` Finding with LensJudge v3 HSC Cross-Validation

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Abstract

We deploy the contaminant-aware ClaudeNet `v3blend8` ensemble and the LensJudge v3 vetting cascade on the final DESI Legacy Surveys data release (DR11). The **DR11-south** (DECam, native *griz*) sweep scores all 5.38×10^7 galaxies with the five-member v2-lean stage and re-ranks the survivors with the eight-member `v3blend8` ensemble; a DR11-native recalibration is required because the deeper DR11 reductions have fatter random-galaxy score tails than DR9, tightening the matched-FPR thresholds (known-lens grade-A recall at the 10^{-4} operating point falls $62\% \rightarrow 41\%$ relative to DR10 — a genuine DR9→DR11 domain shift, not a threshold artifact, so the DR11 survivor list under-recovers grade-A lenses and is not complete). Group-conformal selection remains *power-limited* at $m \approx 5.4 \times 10^7$ (zero certified at $\alpha \leq 0.25$ with an 8M-negative calibration), so the candidate list is the `v3blend8`-ranked top-N, as in the prior campaigns. The decisive step is LensJudge v3's new **HSC PDR3 tier-2**: vetting the top-500 new candidates as a cascade (cheap detection triage → agentic mimic adjudication → high-resolution escalation, $40 total) moves **26** HSC-covered candidates from DESI-ambiguous ($p_{\rm lens} \sim 0.03$–0.6) to **24 HSC tier-2 grade-A/B** under LensJudge's automated grader ($p_{\rm lens}$ up to 0.97; a single automated grader, not a human committee or spectroscopy). An independent SuGOHI cross-match calibrates the engine: `v3blend8` recovers **79** SuGOHI HSC committee lenses into the survivors at 2.7× the survivor-median score, and **15 of the 24** graded A/B are SuGOHI lenses (a sensitivity check — the automated tier-2 grader's precision on the genuinely-new candidates is not established here). The remaining **9** (8 distinct systems) are new to both the DESI lens-finder catalogs and SuGOHI — candidate detections, pending human-expert and spectroscopic follow-up. The honest frame is unchanged from the v3 program: the model recovers known lenses and ranks new candidates the higher-resolution HSC grader rates highly, but *no net-new lens is established from DECaLS pixels alone* (all 9 depend on HSC resolution and remain unconfirmed), and net-new discovery beyond existing high-resolution catalogs is resolution-bounded — the value of the *resolution lever* (now HSC over ∼1,300 deg^2, far wider than Euclid Q1) is to *convert* overlap candidates, not to manufacture discoveries. The **DR11-north** (BASS/MzLS) arm — which needs instrument-native retraining (v3 is DECam-specific) — is fully staged but deferred by a 7-day Perlmutter maintenance.

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1 Premise

The ClaudeNet v3 program concluded that for DECaLS-resolution lens finding the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture: `v3blend8` (the SHIP-gated eight-member ensemble: `effnet_B`, `zoobot_N`, and the v2-hard + v3-mimic-blend versions of `effnet_S2`, `effnet_B3`, `resnet46_C`) rejects common contaminants far better than v2 but is statistically tied with the originals at the hardest real-vs-mimic frontier, which is resolution-bounded (Inchausti Reyes et al., 2025). The productive mode is therefore *DESI pre-screen → high-resolution confirmation in the overlap*. This campaign operationalizes that on the final release (DR11) and exploits the new lever in LensJudge v3: an **HSC PDR3** tier-2 grader (Aihara et al., 2022) ($0.168''$/px, $\sim 0.6''$ seeing) whose footprint (HSC-SSP Wide, $\sim$1,300 deg^2) is $\sim 20\times$ the Euclid Q1 area used previously, vastly widening the confirmable overlap.

2 The DR11-south sweep

DR11-south is the deepest DECam reduction to date: 1,600 sweep files (1.7 TB) yielding a parent of **53,809,040** galaxies after the published Inchausti cuts (`TYPE`$\in\{$SER,EXP,DEV,REX$\}$, NOBS$_{grz} \geq 3$, $m_z < 20$), 51.4M with native i. We shard into 32 footprint-pure, brick-disjoint parts, extract 101-px grz cutouts (100% success, 6.0 TB), and score every galaxy with the five v2-lean members.

DR11-native recalibration (required). The DR9-fit per-member 10^{-4}-FPR thresholds, when transferred to DR11 score-space, are *too tight*: DR11's deeper imaging produces a fatter high-score tail of random galaxies (e.g. the `resnet46_C` isotonic score saturates, its 10^{-4} threshold moving $0.81 \to 1.00$, effectively reducing the union to four members). We recalibrate from the scored population itself (a 8 M-galaxy random NegEval), yielding **95,104** survivors. Re-measuring recall of the published Inchausti 811 against these survivors gives grade-A 41% (37/90), versus 62% for the DR10-recalibrated DR10 sweep — a real DR9→DR11 model domain shift (the DR9-trained members under-score a fraction of DR11 grade-A lenses, *and* the random tail is fatter), not a threshold artifact (the denominator includes north and parent-cut lenses, so 41% is a conservative floor).

Selection. Scoring the survivors with all eight `v3blend8` members and ranking by the mean calibrated score, the top-300 contain **102 known** lenses (recovered into the top of the list — a consistency check) and **198 new** candidates (top-300 `v3blend8` score 0.812–0.939). The discovery set is the **top-500 new** (`v3blend8` score 0.734–0.933); the broader survivor pool has 1,071 survivors above `v3blend8` 0.7 (726 new + 345 already-known).

Certified-FDR (power-limited, reported honestly). Group-conformal selection (Jin and Candès, 2023) with the 8 M NegEval certifies **zero** candidates at $\alpha = 0.05/0.10/0.25$: the per-group p-value floor $1/(n_{\mathrm{cal}}+1) \approx 2.5 \times 10^{-7}$ is $\sim 50\times$ short of what full-m Benjamini-Hochberg needs at $m = 5.38 \times 10^7$ to select even one. This is the same power-floor diagnosed at DR9/DR10 scale — rigorous full-population FDR needs an infeasibly large calibration set — so the candidate list is the `v3blend8`-ranked top-N, not an FDR-certified set. The arithmetic self-test (14/14) and the byte-level scoring-path check pass.

Table 1: DR11-south cascade outcome (500 new candidates, `pass_frac`=0.7).

quantity	value	note
graded (stage-1 triage)	500	\$0.012 each
escalated (agentic stage-2)	350	v2 mimic rubric
reached HSC tier-2	26	HSC-SSP Wide coverage
grade A / B / C / D	27 / 76 / 125 / 272	final (DESI- or HSC-grade)
HSC tier-2 grade-A/B	**24** (19 A, 5 B)	automated; 8 more A are DESI-only
total cost	\$40.38	

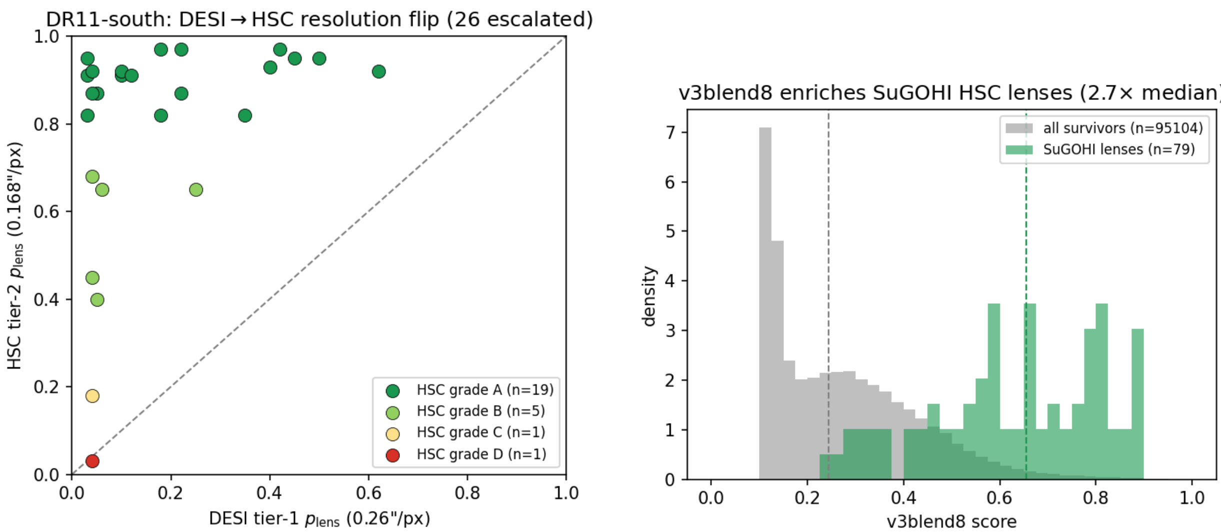


Figure 1: Left: the DESI→HSC resolution flip — the 26 HSC-covered candidates move from DESI-ambiguous (tier-1 $p_{\rm lens}$) to the HSC tier-2 grade (automated grader), 24 landing A/B above the $y = x$ line. Right: `v3blend8` enriches the independent SuGOHI HSC lenses to 2.7× the survivor-median score (median lines dashed).

3 LensJudge v3 cascade vetting + HSC cross-validation

The DESI viewer exposes no public DR11 layer (and is unreachable from our host), so we stage the 500 candidates' *grz* cubes from Perlmutter as on-disk FITS, ensuring LensJudge grades the exact pixels the CNN scored. We then run the B7 deploy cascade: a loop-free `direct` detection triage on all 500 (∼\$0.012 each), escalating the top 70% by rank to the agentic v2 mimic-adjudication grader, which itself escalates to **HSC PDR3 tier-2** wherever HSC covers the position. Total cost **\$40.4** (campaign total \$43 of a \$250 budget).

The resolution lever converts. Every tier-2 grade-A/B candidate shows a large DESI→HSC flip: median tier-1 $p_{\rm lens} \approx 0.1$ rising to tier-2 0.82–0.97 (e.g. 0.03 → 0.95, 0.04 → 0.92). At DECaLS 1″ seeing these are ambiguous (lrg+companion ≈ lens); at HSC 0.6″ the arc/ring morphology resolves — the flip is the grader's response to resolution, not an observational confirmation.

Table 2: The 9 genuinely-new HSC tier-2 grade-A/B candidate detections (8 distinct systems; new to DESI lens-finders *and* SuGOHI; automated grader, follow-up pending). p_1, p_2 = LensJudge tier-1 (DESI) / tier-2 (HSC) lens probability.

name	RA	Dec	grade	$p_1 \to p_2$	`v3blend8`
s_340513_3688	16.332	+1.749	A	0.22 → 0.97	0.77
s_327526_9059	9.722	−0.409	A	0.04 → 0.92	0.74
s_351303_461	194.534	+3.536	A	0.62 → 0.92	0.89
s_345385_1515†	154.312	+2.489	A	0.10 → 0.92	0.81
s_340971_2860	130.857	+1.784	A	0.12 → 0.91	0.76
s_345385_1514†	154.312	+2.488	A	0.05 → 0.87	0.86
s_326089_1785	10.288	−0.730	A	0.03 → 0.82	0.83
s_325163_2981	138.848	−0.923	A	0.18 → 0.82	0.74
s_318043_1037	158.789	−2.304	B	0.04 → 0.45	0.86

† s_345385_1514/1515 are adjacent (∼1″) detections of the same system, so the 9 detections are **8 distinct candidate systems** (8 A-detections / 7 distinct grade-A + 1 grade-B).

Independent SuGOHI validation. Against the SuGOHI/HOLISMOKES HSC committee lens catalog — an *independent* sample with expert grades and spectroscopic redshifts (Sonnenfeld et al., 2018) — `v3blend8` recovers **79** lenses into the 95,104 survivors at a median `v3blend8` score 2.7× the survivor median (27 in the top-500), and **15 of the 24** tier-2 grade-A/B candidates are SuGOHI lenses. These 15 are a *sensitivity* check — the automated HSC grader recovers known committee lenses as grade-A/B — but they do not establish its *precision* on the genuinely-new candidates: no negative control was run through tier-2 and the certified-FDR is power-limited (§2), so the false-positive rate of the new set is unquantified.

The candidate catalog. Of the 24 tier-2 grade-A/B candidates, **0** are in the DESI lens-finder catalogs (by construction — the 500 were the new set after removing the 102 known), 15 are SuGOHI cross-matches, and **9** (8 distinct systems) are new to both the DESI finders and SuGOHI (Table 2) — the campaign's candidate detections. The honest caveats: LensJudge's HSC grading is automated (a single Claude grader, not a human committee, no spectroscopy), and "new" means new to the DESI finders and SuGOHI, not to every HSC lens search. They are candidates warranting human-expert and spectroscopic follow-up, not confirmed lenses.

Active-learning loop. The cascade's agent-confirmed mimics (grade-D + named contaminant) export to **291** hard negatives in the ClaudeNet mimic-bank schema — `lrg_companion` 243-dominant, confirming that the high-`v3blend8` new frontier is lrg+companion-dominated at DECaLS resolution (exactly the population the HSC tier-2 rejects). Folding these into the bank and a model re-iteration are Perlmutter GPU work, deferred (below).

4 DR11-north: staged, maintenance-deferred

v3 is DECam-specific — it systematically under-scores BASS/MzLS north imaging — so the north arm requires instrument-native retraining. We built the full north parent (1.16×10^7 galaxies, *grz* only, no *i*), an 8-part manifest, and extracted **523** north-native training positives (297 BASS/MzLS curated + 226 non-leaked catalog lenses; 86 clean held-out; 33 v1 leaks tracked); the retrainer is patched to fine-tune the three swappable members from their `_b50` checkpoints on a north-native mimic blend. A full-system Perlmutter maintenance (2026-06-17 to 06-24) began mid-campaign before the north extraction could be scheduled, so the north sweep + retrain + validation gates are **deferred to post-maintenance**; all inputs are staged for a clean resume.

5 Conclusion

On the final DESI Legacy Surveys release, the contaminant-aware ClaudeNet finder plus the LensJudge v3 HSC cascade is a working DESI×HSC cross-validation engine on the DECam-south footprint: it scores 5.4×10^7 galaxies, recalibrates honestly to the DR11 domain shift (with a real ~34% relative grade-A recall loss vs DR10 — the survivor list is incomplete), and — through the new HSC tier-2 over a 20×-wider footprint than Euclid Q1 — moves 24 DESI-ambiguous candidates to HSC tier-2 grade-A/B under LensJudge's automated grader, of which 15 are SuGOHI committee lenses (a sensitivity check) and 9 (8 systems) are new to the DESI finders and SuGOHI. The certified-FDR is power-limited (zero) at full-population scale, the tier-2 grader is automated (no human committee or spectroscopy, precision on the new set unquantified), and *no net-new lens is established from DECaLS pixels alone* — all 9 candidate detections depend on HSC resolution and remain unconfirmed pending follow-up. This is the same honest ceiling the v3 program established — the resolution lever *converts* overlap candidates rather than manufacturing discoveries — now measured on DR11 with a much wider confirmable overlap. The north footprint, requiring instrument-native retraining, is staged for completion after the Perlmutter maintenance window.

References

Hiroaki Aihara et al. Third data release of the hyper suprime-cam subaru strategic program. *Publications of the Astronomical Society of Japan*, 74(2):247–272, 2022.

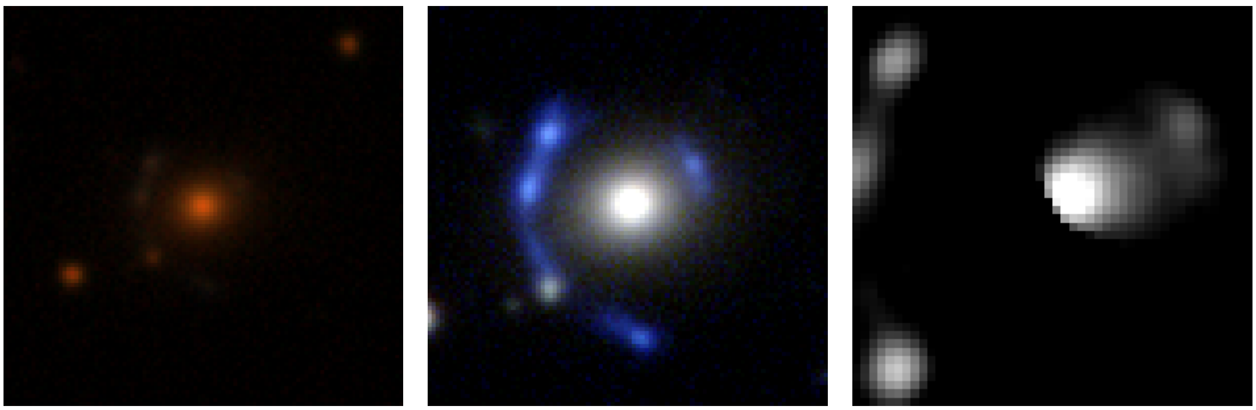
J. Inchausti Reyes et al. Strong lens discovery in DR10 with two deep-learning architectures. *arXiv:2508.20087*, 2025.

Ying Jin and Emmanuel J. Candès. Selection by prediction with conformal p-values. *JMLR*, 24, 2023.

Alessandro Sonnenfeld et al. Survey of gravitationally-lensed objects in hsc imaging (sugohi). i. *Publications of the Astronomical Society of Japan*, 70:S29, 2018.

A The new candidate systems at DESI and HSC resolution

The figures below show the **8 distinct new candidate systems** (the 9 detections of Table 2; the `s_345385` pair is one system). Each is rendered as: the ClaudeNet-seen DESI *grz* cutout (0.262″/px, Lupton RGB — what the finder scored, ambiguous at DECaLS resolution); the HSC PDR3 *grizy* cutout (0.168″/px — the same view LensJudge's tier-2 grader inspected); and the HSC lens-light-subtracted view. The DESI→HSC contrast is the campaign's mechanism made visual: the finder ranks these high from ambiguous DECaLS pixels, and the resolution lever reveals the arc/ring morphology (clearest in `s_340513_3688`; fainter and correspondingly less certain in others). These are *automated-grader candidates, not confirmed lenses* — they warrant human-expert and spectroscopic follow-up.

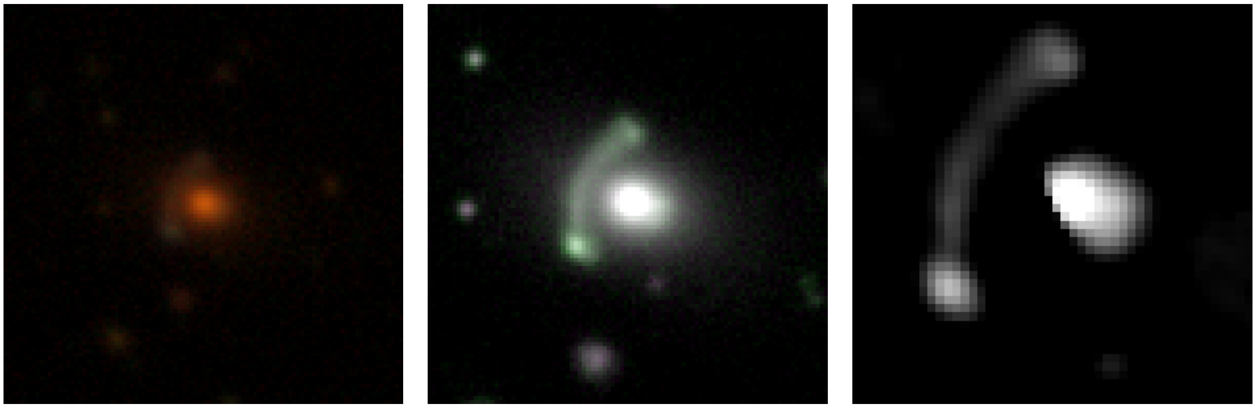


DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 2: `s_340513_3688` (RA 16.3319, Dec +1.7490) — HSC tier-2 grade **A**, p_{lens} 0.22→0.97, `v3blend8` 0.77.

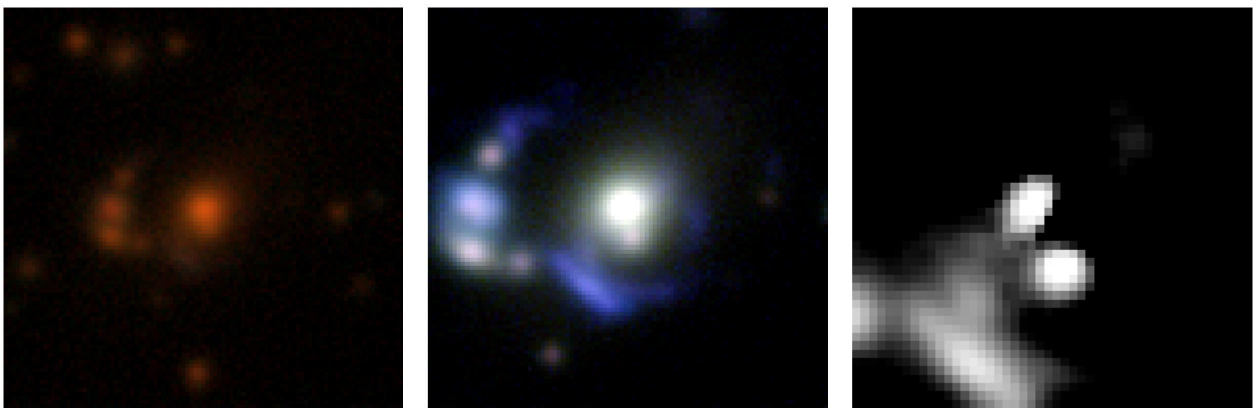
DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 3: `s_327526_9059` (RA 9.7222, Dec -0.4093) — HSC tier-2 grade **A**, p_{lens} 0.04→0.92, `v3blend8` 0.74.



DESI *grz* (0.262″/px) | HSC *grizy* (0.168″/px) | HSC lens-light subtracted

Figure 4: `s_351303_461` (RA 194.5345, Dec +3.5358) — HSC tier-2 grade **A**, p_{lens} 0.62→0.92, `v3blend8` 0.89.

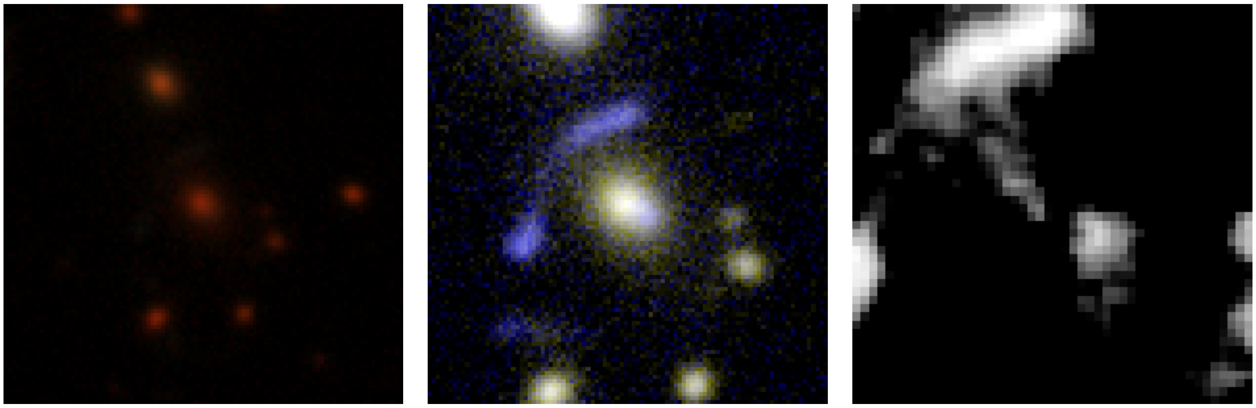


DESI *grz* (0.262″/px) | HSC *grizy* (0.168″/px) | HSC lens-light subtracted

Figure 5: `s_345385_1515` (RA 154.3116, Dec +2.4885) — HSC tier-2 grade **A**, p_{lens} 0.10→0.92, `v3blend8` 0.81. (= 8th detection s_345385_1514, the adjacent ∼1″ pair, omitted)

DESI *grz* (0.262″/px) | HSC *grizy* (0.168″/px) | HSC lens-light subtracted

Figure 6: `s_340971_2860` (RA 130.8571, Dec +1.7840) — HSC tier-2 grade **A**, p_{lens} 0.12→0.91, `v3blend8` 0.76.

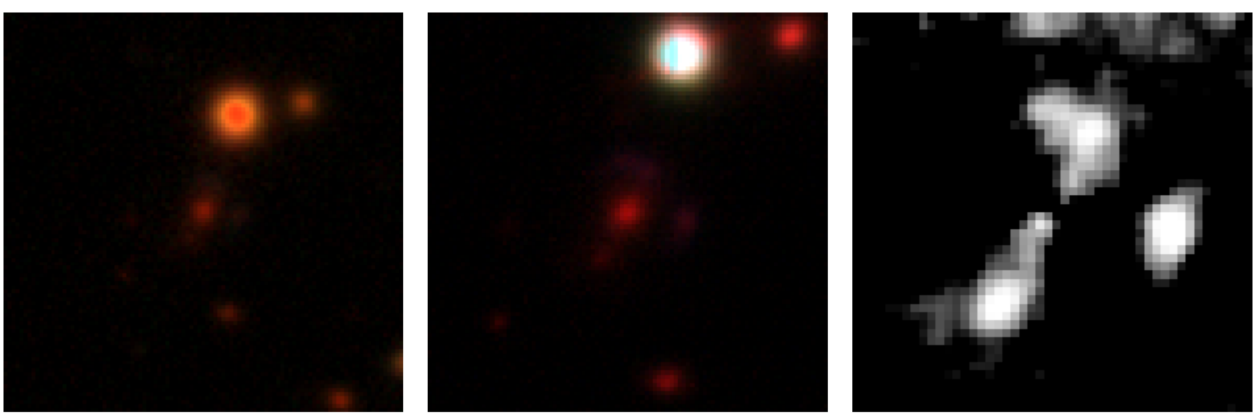


DESI *grz* ($0.262''$/px) HSC *grizy* ($0.168''$/px) HSC lens-light subtracted

Figure 7: `s_326089_1785` (RA 10.2876, Dec -0.7303) — HSC tier-2 grade **A**, $p_{\rm lens}$ $0.03 \rightarrow 0.82$, `v3blend8` 0.83.

DESI *grz* ($0.262''$/px) HSC *grizy* ($0.168''$/px) HSC lens-light subtracted

Figure 8: `s_325163_2981` (RA 138.8482, Dec -0.9226) — HSC tier-2 grade **A**, $p_{\rm lens}$ $0.18 \rightarrow 0.82$, `v3blend8` 0.74.



DESI *grz* ($0.262''$/px) HSC *grizy* ($0.168''$/px) HSC lens-light subtracted

Figure 9: `s_318043_1037` (RA 158.7893, Dec -2.3037) — HSC tier-2 grade **B**, $p_{\rm lens}$ $0.04 \rightarrow 0.45$, `v3blend8` 0.86.